2017 Ph H1 Q9

Section: Particles and Waves

Topic: Forces on Charged Particles

Brief summary of the question

A clean zinc plate is illuminated.

The radiation is replaced by **higher frequency** light

at the **same irradiance**.

Predict the changes in:

(i) maximum kinetic energy of emitted photoelectrons

(ii) number of photoelectrons per second

Worked solution

- Photon energy: $E = hf$.

Higher f ⇒ higher energy **per photon**.

- Photoelectric equation: $K_{\max} = hf - \phi$.

ϕ is constant for the metal.

If f increases, $K_{\max}$ **increases.**

- Irradiance is fixed (same power per area).

Each photon now has **more** energy,

so **fewer** photons arrive per second.

Fewer photons ⇒ **fewer electrons** emitted per second.

Final answer

D — $K_{\max}$ increases; number per second decreases.

Revision tips

- At constant irradiance: ↑f ⇒ ↑$K_{\max}$, ↓photon rate.
- ↑Irradiance changes the **rate**, not $K_{\max}$.
- ↑Frequency changes $K_{\max}$, not the brightness.